Code using RAW sockets to implement packet sniffing.

EXP. NO : 14
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Aim :

To capture and analyse network packets and display that protocol's source IP address and destination IP address.

Program :

```
from scapy.all import sniff
from scapy.layers.inet import IP, TCP, UDP, ICMP

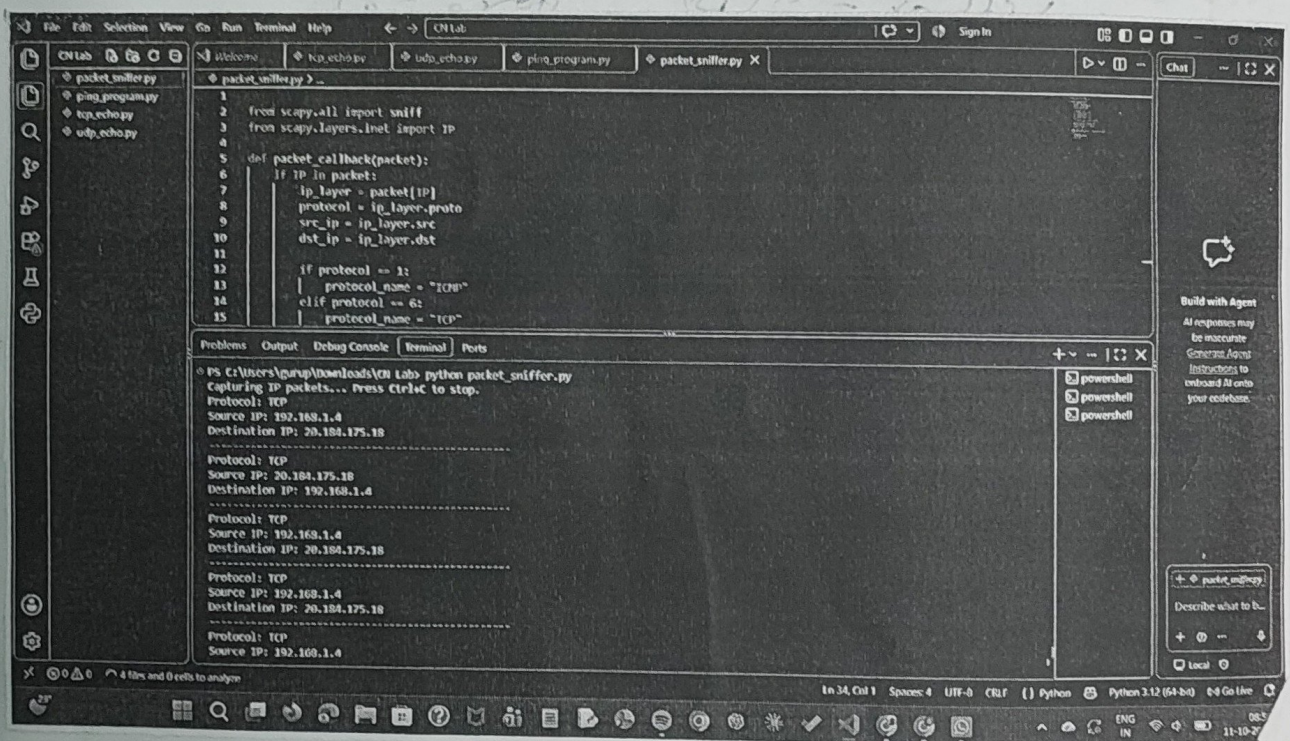
def packet_call_back(packet):
    if IP in packet:
        IP_layer = packet[IP]
        protocol = IP.layer.proto
        src_ip = ip_layer.src
        dest_ip = ip_layer.dst
# determine the protocol
protocol_name = ""
if protocol == 1:
    protocol_name = "ICMP"
elif protocol == 6:
    protocol_name = "TCP"
elif protocol == 17:
    protocol_name = "UDP"
else:
    protocol_name = "unknown protocol"
```

```
# print packet details
print(f"Protocol: {proto_name}")
print(f"Source IP: {src_ip}")
print(f"Destination IP: {dst_ip}")
print(f"{'-'*50}")

# Capture packets on the default network interface
sniff(iface="wifi", prn=packet_callback,
      filter="IP", store=0)

if __name__ == "__main__":
    print("Capturing IP packets... press
                ctrl + c to stop")

    sniff(
        iface="Name",
        prn=packet_callback
        filter="up"
        store=False
        )
```



Result :

The packet-sniffing program was successfully implemented.